

Roger Richard Demello

Software Engineer, AI/ML | India | B.Tech 2026 | Immediate joiner | Fully remote, any timezone

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PROFESSIONAL SUMMARY

Software engineer shipping **machine learning** and **generative AI** systems end to end – **Python** and **TypeScript** services over **PostgreSQL**, **Docker** delivery through **CI/CD**, **3 production applications** for **200+ users** at **99.5% uptime**.

TECHNICAL SKILLS

Programming Languages: Python, Java, TypeScript, JavaScript, SQL, C++, C, Bash, YAML

Machine Learning: Scikit-learn, PyTorch, XGBoost, LightGBM, anomaly detection, NLP, computer vision

Generative AI: Retrieval-Augmented Generation (RAG), LangChain, LangGraph, multi-agent systems, FAISS

Backend and APIs: FastAPI, Flask, Express.js, Node.js, REST APIs, microservices, React, Next.js

Data and Cloud: PostgreSQL, Redis, Pandas, NumPy, MLflow, Docker, AWS, CI/CD, GitHub Actions, MLOps

CS Fundamentals: data structures, algorithms, system design, object-oriented programming, pytest

WORK EXPERIENCE

Innovun Global

August 2026 – Present

AI Engineer Intern

- Build the retrieval pipeline and prompts behind a live **RAG** agent for **Cruz Roja Mexicana** (Red Cross, Mexico).
- Integrate **two external platform APIs** across **3 channels** into one consistent schema behind a live service.
- Verify production defects from first report through fix and post-release check, with client **stakeholders**.
- Author **technical documentation** of root causes and fixes in clear written English, so a defect is solved once.

AI LifeBOT (B2B Startup)

January 2026 – June 2026

AI Engineer Intern

- Engineered **3 production applications** and backend services for **200+ users**, raising task efficiency **40%**.
- Reduced query and response **latency 35%** by profiling an inherited codebase across the full request path.
- Shipped **5+ features** end to end, compressing delivery and validation cycles **50%** with cross-functional teams.
- Automated **pytest** suites inside **CI/CD** pipelines and reviewed teammate code in an **Agile** team.

CFM, RCOEM

May 2025 – July 2025

Machine Learning Research Intern

- Trained a clinical **classification** model on **1,000+** health records, reaching **87% accuracy** on held-out data.
- Engineered measurable **features** in **Python** and **Pandas** from raw, inconsistent records before training.
- Raised result reliability **25%** through **cross-validation** across multiple splits, not one sample.
- Standardized **data pipelines** into **tested, reusable components**, cutting development time **30%**.

PROJECTS

music-recsys | *Python, PyTorch, LightGBM, FastAPI, MLflow, Redis, Kafka, Prometheus, Grafana, Docker* | [GitHub](#)

- Delivered a two-stage **ranking and recommendation** system – two-tower **embeddings** into ANN retrieval, then **LightGBM**.
- Scheduled retrain, embedding-refresh and candidate-precompute jobs against an **MLflow** model registry.
- Simulated **A/B tests** across ranking variants, with **Prometheus** and **Grafana** tracking model and serving metrics.
- Abstracted cache, event bus, feature store and ANN index behind interfaces, scaling local to **Kafka** by config.

sentinelops | *Python, FastAPI, uvicorn, Scikit-learn, statsmodels, NetworkX, React, Vite, TypeScript, Supabase* | [GitHub](#)

- Architected an incident prediction platform behind **FastAPI** services and a **React** and **TypeScript** console.
- Detected anomalies with **IsolationForest** over generated telemetry, modelling service dependencies in **NetworkX**.
- Automated **Slack** and **PagerDuty** alerting with simulated self-healing and structured audit logging per action.
- Designed optional hookup to real Kubernetes and webhook infrastructure without changing the core service contracts.

DealSentry | *TypeScript, React, Express.js, Prisma ORM, PostgreSQL, Node.js, OpenAI API, Puppeteer* | [Live](#) | [GitHub](#)

- Built an enterprise proposal platform on **Express.js**, **Prisma ORM** and **PostgreSQL** with **role-based access control**.
- Automated a compliance **rules engine** scoring pricing, legal and structural risk before signature rather than after.
- Integrated **Salesforce**, **HubSpot**, **Gmail** and Google Drive, with document upload, parsing and PDF generation.
- Routed approvals with **SLA tracking** and audit logging, so every review step stays attributable and time-bound.

EDUCATION

Shri Ramdeobaba College of Engineering and Management (RCOEM)

2022 – 2026

B.Tech, Electronics & Communication Engineering

GPA: 8.90/10

Minor: **Artificial Intelligence & Machine Learning**

GPA: 9.67/10

CERTIFICATIONS AND AWARDS

AWS Certified Cloud Practitioner – Amazon Web Services (October 2025)

Hackathon Finalist – Paytm x Sarvam x Logitech AI National Hackathon

Second Place – ByteSize Sage AI National Hackathon